

Examination paper for MOL3000 and MOL3100 - Introduction to Molecular Medicine

Date: 08.12.2025

Time: 15.00-19.00

Course contact: Bodil Kavli

Present at the exam location: No

Permitted examination support material: D

No printed or hand-written support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

Read the questions carefully and make your own assumptions. In your answers, explain clearly what assumptions you have made and how you have understood or limited the assignment

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website "Explanation of grades and appeals".

SPECIFIC INFORMATION

No paper drawings:

This exam does not include hand drawings. If you receive hand drawing sheets, this is by mistake. You will not be able to submit the sheets, and they will not be graded.

Weighting: Maximum score is 100 marks. Weighting is indicated on each question.

Withdrawing from the exam:

If you wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Access to your answers:

After the exam, you can find your answers under previous tests in Inspira. Be aware that it may take a working day until any hand-written material is available in "previous tests".

Essay Q1 (4 marks)

The COVID-19 pandemic accelerated the approval of several vaccines against the SARS-CoV-2 corona virus, including mRNA vaccines, which became a gamechanger to overcome the pandemic.

Explain the principle of mRNA vaccines and indicate how the synthetic mRNA is optimized to increase antigen production and prevent unwanted inflammatory responses in the host.

Answer:

In mRNA vaccines, the mRNA encodes a harmless piece of the virus—usually the spike protein in the case of COVID-19.

The mRNA is packaged in lipid nanoparticles (tiny fat bubbles) to protect it and help it enter cells.

Once injected, these particles deliver the mRNA into muscle cells near the injection site.

Inside the cell, the mRNA is read by ribosomes, which produce viral protein that acts as an antigen. The immune system responds by activating T-cells, produce antibodies and creates immunological memory (as other vaccines)

The mRNA is optimized by modifications to the bases (e.g. pseudouridine)

The Nobel Prize in Physiology or Medicine 2023

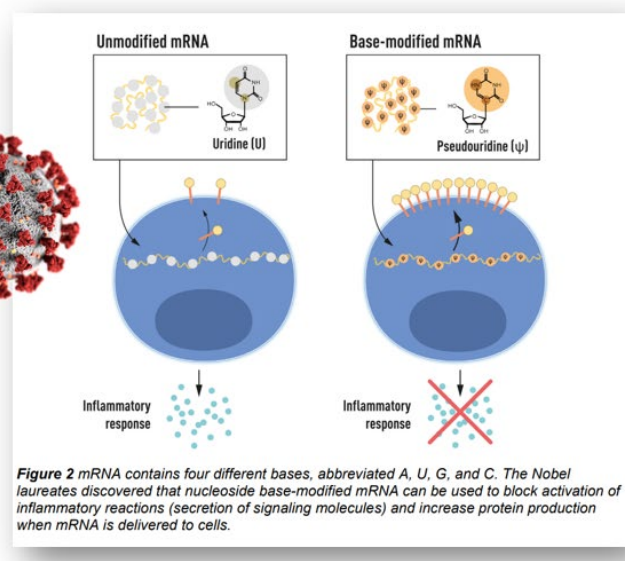
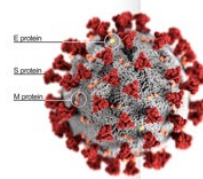


© Nobel Prize Outreach. Photos:
Clément Morin
Katalin Karikó
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The Nobel Prize in Physiology or Medicine 2023 was awarded jointly to Katalin Karikó and Drew Weissman "for their discoveries concerning nucleoside base modifications that enabled the development of effective mRNA vaccines against COVID-19"



Essay Q2 (8 marks)

Development of new drugs based on recombinant expression of proteins is important in molecular medicine.

- A. List at least four main classes of recombinant protein drugs (4 marks).
- B. Give examples of at least two commonly used production organisms and indicate their advantages and disadvantages as hosts for recombinant protein drugs (4 marks)

Answer:

A. Hormones, coagulation factors, enzymes, cytokines, vaccines, antibodies, peptides

B.

TABLE 10.3 Comparison of different prokaryotic and eukaryotic host organisms for the recombinant expression of proteins.		
Host organism	Advantages	Disadvantages
Bacteria (<i>E. coli</i>)	- Well-characterized - Great choice of expression vectors - High yields - Low cost and easy handling	- No posttranslational modifications - Formation of inclusion bodies - May be immunogenic - May contain bacterial impurities, for example, endotoxins derived from Gram-negative bacteria
Yeast	- No endotoxins detectable - Fermentation is well-established and comparably cheap - Allows glycosylation and the formation of disulfide bonds	- Glycosylation is not identical to the pattern observed in mammalian cells - Protein expression is more difficult to control than in <i>E. coli</i>
Insect cells	- Posttranslational modifications (e.g. glycosylation) similar to that in mammalian cells - Biologically safe	- More demanding culture conditions - Mechanism of glycosylation is not fully understood - Function and immunogenicity may be problematic
Mammalian cells	Highest level of posttranslational protein processing, particularly glycosylation patterns, and formation of disulfide bonds	- Demanding and expensive cell culture - Lower yields than in microorganisms
Transgenic plants and animals	- Posttranslational modifications - Large-scale production	Very complex and time-consuming generation of transgenic lines

Essay Q3 (7 marks)

DNA is the carrier of genetic information. However, it has limited chemical stability and various DNA repair mechanisms are critical for genome maintenance.

- A. Describe the main steps and enzymes in the base excision repair (BER) pathway. (2 marks)
- B. List three sources to uracil in DNA and indicate their mutagenic potential. (3 marks)
- C. In what way is uracil in DNA a key intermediate in the generation of antibody diversity? (2 marks)

Answer:

A)

- Excision of damaged/wrong base by DNA glycosylase,
- cutting of DNA strand by AP-endonuclease,
- insertion of new nucleotide (removal of deoxyribose-phosphate) by DNA polymerase
- strand ligation by ligase

B)

1. Incorporation of uracil instead of thymine, U:A not miscoding/mutagenic
2. Spontaneous deamination of cytosine, U:G, 100 % miscoding/mutagenic
3. Enzymatic deamination of cytosine by deaminase, U:G, 100 % miscoding/mutagenic

C)

Uracil is intermediate in somatic hypermutation and class switch recombination to generate antibodies with high affinity and of different isotypes

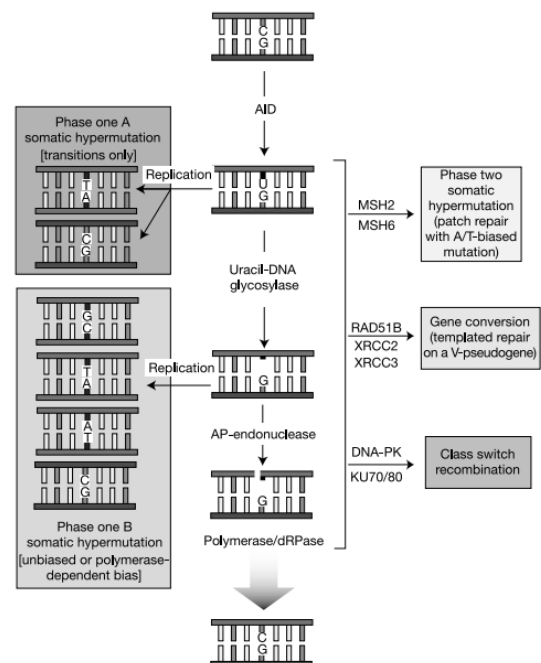


Figure 1 DNA deamination model of immunoglobulin gene diversification. For details,

Essay Q4 (6 marks)

A child presents with recurrent respiratory infections. Genetic testing identifies a homozygous Gly542X nonsense mutation in the *CFTR* gene (1480 amino acids) and confirms cystic fibrosis (CF).

- A. Describe the genetic basis and inheritance pattern of CF. (1 marks)
- B. How can the CF diagnosis explain the clinical symptoms in the patient? (2 marks)
- C. Trikafta, an effective combination therapy that stimulates transport of CFTR to the cell membrane and activates its function, has revolutionized the treatment for CF patients with the common Phe508 deletion mutation. Would “Trikafta” be an effective treatment for this patient (Gly542X)? Argue for your answer. (3 marks)

Answers:

- A. It is inherited in an autosomal recessive pattern which means that a person must inherit two copies of the disease allele to develop the disease.
- B. CFTR protein regulates chloride and water transport across cells. Mutations lead to a dysregulated flow of ions which results in a thick, sticky mucus in the lungs, causing respiratory infections.
- C. The 542X mutation results in an early stop codon (position 542 of total 1480) – thus, no CFTR protein is produced. Therefore, Trikafta would not work on this mutation.

Essay Q5 (7 marks)

Cancer stem cells are specialized types of cancer cells found within tumors.

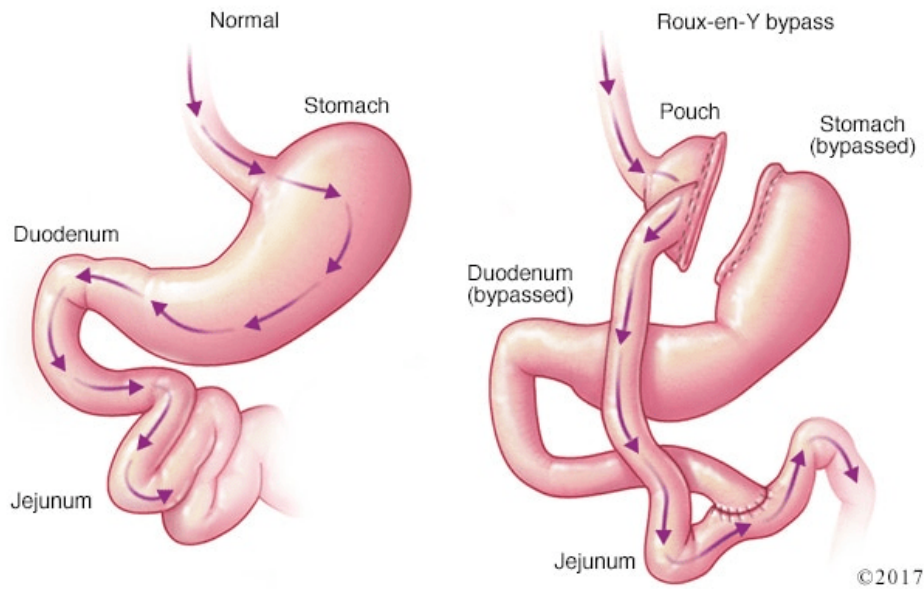
- A. Which three key advantages are cancer stem cells responsible for in the tumour? (3 marks)
- B. What are the two functionally distinct subsets of cancer stem cells, and what is an important difference between them? (2 marks)
- C. Name and describe two modes of stem cell division? (2 marks)

Answers:

- A. seeding tumours, resistance to therapy and tumour heterogeneity
- B. Two functionally distinct subsets are stationary and migratory cancer stem cells. An important difference is that stationary cancer stem cells are embedded in the primary tumour, while migratory cancer stem cells impart metastatic activity.
- C. Asymmetric and symmetric. Asymmetric division results in one daughter stem cell and one daughter cell that is destined to differentiate. Symmetric division generates two identical cells (either two stem cells or two differentiated cells).

Essey Q6 (6 marks)

Bariatric surgery (Roux-en-Y bypass) is known to induce not only bodyweight reduction but also type 2 diabetes (T2D) remission.



- A. What is current understanding of mechanism behind the T2D remission after such surgery?
- B. How can this knowledge be used in development of medicine for T2D?

Answer:

- A. *Due to changes in anatomy by the surgery, ingested food reaches rapidly the small intestine where L cells are located, leading to an exaggerated release of GLP-1 and PYY, and then GLP-1 and PYY stimulate the beta-cells in pancreas, leading to insulin release.*
- B. *Potential medicines for T2D may include molecules that mimic GLP-1 and PYY. Since GLP-1 has a short half-life (1.5 – 5 min) due to dipeptidyl peptidase-4 (DPP4), DPP4 inhibitors can be considered for treatment of T2D (In addition, new research indicated that “inactive GLP-1 (9-36)” might have the incretin effect too).*

Essay Q7 (5 marks)

Each winter comes with a new influenza season. The influenza virus evolves via two mechanisms, termed antigenic drift and antigenic shift.

Explain the principles of these two mechanisms and the potential consequences of both.

Sensor guide:

Influenza viruses evolve through antigenic drift and antigenic shift, both of which alter the surface proteins hemagglutinin (HA) and neuraminidase (NA)—the main targets of the immune system.

Antigenic Drift: Antigenic drift is a gradual process caused by point mutations in the viral RNA, especially in HA and NA genes. These mutations arise due to the lack of proofreading by the viral RNA polymerase. Over time, they lead to minor antigenic changes that reduce antibody recognition, allowing the virus to evade immunity and cause seasonal epidemics. This necessitates annual updates to influenza vaccines.

Antigenic Shift: Antigenic shift is a sudden, major change that occurs when two different influenza A viruses infect the same host cell and reassort their segmented genomes. This can produce a virus with a novel HA and/or NA subtype, to which humans have little or no immunity. Antigenic shift can lead to pandemics, as seen in 1918, 1957, 1968, and 2009.

Essay Q8

Atherosclerosis is a major cause for cardiovascular diseases.

- A. What is atherosclerosis? Include the biological processes involved and how they affect blood vessels in your answer. (3 marks)
- B. Atherosclerosis can lead to myocardial infarction or stroke. Describe the sequence of events that results in these outcomes. (2 marks)
- C. Why are the heart and brain particularly vulnerable to damage caused by atherosclerosis? Include the physiological reasons and consequences in your answer. (2 marks)

Answer:

- A. Atherosclerosis is the formation of lipid-rich plaques in arteries, which will eventually lead to a blockade of blood vessels or to the formation of a blood clot.
- B. Rupture of plaques (either through erosion or rupture) leads to the local coagulation or clotting of the blood due to activation of the coagulation cascade upon contact of blood with the core of the plaque. Coagulation blocks the blood supply in the heart or in the brain eventually causing myocardial infarction or stroke.
- C. The heart and brain need a constant supply of oxygen and nutrients. Atherosclerosis can block blood flow to these organs, leading to heart attacks (if coronary arteries are affected) or strokes (if brain arteries are blocked). Their high energy demand and sensitivity to reduced blood flow make them especially vulnerable. There is very limited regeneration capacity of these organs

MCQs

1. The non-protein-coding fraction (%) of genomes correlates with complexity of organisms.

In the human genome this part constitutes

~75%

~50%

~90%

~98%

2. DNA methylation is involved in regulation of gene expression.

Which of the following best explains how DNA methylation contributes to gene silencing?

It directly blocks RNA polymerase binding to the promoter.

It recruits methyl-binding proteins that compact chromatin.

It increases histone acetylation, preventing nucleosome assembly.

It promotes transcription factor phosphorylation

3. Histone acetylation is involved in gene regulation.

How does histone acetylation regulate transcription?

By condensing chromatin and preventing transcription factor access

By recruiting DNA methyltransferases to CpG islands

By neutralizing positive charge of Lysine, weakening histone–DNA interactions

By stabilizing nucleosome–nucleosome contact

4. DNA methylation marks can be removed.

Active DNA demethylation differs from passive demethylation primarily because it requires DNA replication to dilute methylation marks

is mediated by DNA repair–dependent enzymatic processes

involves the incorporation of unmethylated cytosines during replication

occurs only during mitosis

5. The genotype is the combination of alleles at a particular DNA segment. A single nucleotide polymorphism with 3 alleles (A, C, T) give

3 possible genotypes

4 possible genotypes

5 possible genotypes

6 possible genotypes

6. You send your DNA to genome sequencing and get back a report of all your single nucleotide variants (SNVs) with allele frequencies in the European population.

The majority of your SNVs would have an allele frequency in the population that is

- higher than 5%
- lower than 5%
- between 0.5-5%
- lower than 0.5%

7. To generate patient-specific models, researchers are continuously developing new procedures. Important progress has been made by the introduction of two techniques: transdifferentiation and direct reprogramming.

Which statement correctly distinguishes transdifferentiation from direct reprogramming?

Direct reprogramming uses lineage-specific transcription factors to push cells directly into the target lineage.

Transdifferentiation involves a temporary pluripotent-like stage before lineage specification.

Direct reprogramming reverts cells fully to a pluripotent state before differentiation.

Transdifferentiation bypasses pluripotency and directly converts one differentiated cell type into another.

8. Organoids are emerging as highly relevant physiological and disease models. Which method is often used to assess organoid maturity at the molecular level?

- A RT-PCR analysis
- B Size measurement
- C Karyotype testing
- D Proliferation analysis

9. Organoids are revolutionizing the molecular medicine research.

Which characteristics best defines organoids?

- A They are derived from fully differentiated cells and remain in 2D culture
- B They originate from stem cells and can self-organize into 3D structures
- C They require external scaffolds for any differentiation to occur
- D They cannot mimic tissue-specific architecture

10. Hematopoietic stem cells are type of adult stem cells that can self-renew and differentiate into more specialized cells.

Where do hematopoietic stem cells primarily reside, and what regulates their fate?

- A In spleen, regulated by macrophages
- B In peripheral blood, regulated by immune cells
- C In lymph nodes, regulated by T cells
- D In bone marrow niche, regulated by endothelial cells

11. Diagnostic criteria for diabetes usually include fasting plasma levels of glucose, two hours after glucose load and HbA1c.

What is the molecular mechanism underlying HbA1c?

- A Hyperglycemia in red and white blood cells
- B Molecular binding of glucose to hemoglobin**
- C Glucose toxicity in red blood cells
- D Diabetes complication in red blood cells

12. Type 1 diabetes (T1D) starts with genetic predisposition in combination with external triggers to induce auto-immune reaction and progresses with loss of beta-cell mass.

How to estimate the beta-cell mass in patients?

- A Plasma glucose level
- B Plasma HbA1c level
- C Plasma C-peptide level**
- D Plasma insulin level

13. Pancreas has both exocrine and endocrine functions.

Which cell type produces somatostatin?

- A Beta cells
- B Alpha cells
- C Delta cells**
- D Nerve fiber

14. A molecular diagnostic panel detects multiple pathogens in a respiratory sample from a patient with pneumonia. What is the main advantage of using such a panel compared to traditional culture?

- A) Faster identification of pathogens, enabling earlier targeted therapy**
- B) Lower cost per test
- C) Higher specificity for all pathogens
- D) Ability to detect only bacteria, not viruses

15. What is the main advantage of mRNA vaccines compared to traditional vaccine technologies?

- A) They stimulate a stronger antibody response than any other vaccine type
- B) They can be produced more rapidly and adapted quickly to emerging pathogens**
- C) They do not require any cold storage, making distribution easier
- D) They eliminate the need for booster doses in all populations

16. Two patients are infected with the same bacterial species, but only one develops severe disease. What is the most likely explanation?

- A) The bacteria in one patient had a larger core genome
- B) One patient had a higher bacterial inoculum and weaker immune response**
- C) One patient had received a vaccine against a different pathogen
- D) The bacteria in one patient were more sensitive to antibiotics

17. A bacterial strain is found to carry genes for both toxin production and antibiotic resistance. What is the most likely explanation for this combination?

- A) The genes are part of the bacterial core genome
- B) The genes were acquired through spontaneous mutation
- C) **The genes are located on a mobile genetic element such as a plasmid**
- D) The genes are only expressed in the presence of antibiotics

18. Inherited mutations in proto-oncogenes and tumor-suppressor genes found in certain types of cancers that run in families. Why are inherited mutations more common in tumor suppressor genes than in proto-oncogenes?

Tumor suppressor mutations cannot affect cancer growth in absence of active oncogenes

The selection against mutation in one allele is lower in tumor suppressor genes than proto-oncogenes

Mutations in tumor suppressor genes have low penetrance

Mutations in proto-oncogenes are not compatible with life

19. Multiple kinase-inhibitors used in cancer treatment targets receptor tyrosine kinases (RTKs); however, their inhibitory effect is often temporary. What is the main reason for this?

Mutations inactivating the binding of the inhibitor to the RTK (gatekeeper mutations)

Crosstalk between pathways and re- organization of the networks

Cancer cells upregulate drug efflux pumps

Cancer cells alter their ability to break down the kinase inhibitor

20. Many cancer treatments target rapidly proliferating cells. Which of the following treatments is targeting cellular features independent of rapid proliferation?

RTK inhibition

Chemotherapy

Immune check point therapy

PARP inhibition

21. Senescence is a new Hallmark of Cancer.

What is senescence?

A form of programmed cell death

A state of permanent cell cycle arrest

A temporary pause in cell division

A type of immune cell activation

22. PSI-BLAST is regarded as an efficient improvement when searching for sequence similarity

Which of the following describes the key improvement introduced by PSI-BLAST?

PSI-BLAST optimize multiple global alignments by adjusting the Smith Waterman threshold

PSI-BLAST find similar sequences from different databases to remove less significant profiles

PSI-BLAST utilize sequence profiles from a multiple alignment

PSI-BLAST weights different profiles against the most likely local amino-acid match.

23. Noncoding RNAs are involved in many biological processes.

Which of the following RNA species is involved in the silencing of transposons?

micro RNA (miRNA)

small nuclear RNA (snRNA)

small nucleolar RNA (snoRNA)

piwi-interacting RNA (piRNA)

24. Aptamers are useful in therapeutics and diagnostics.

How do aptamers work?

By downregulating protein expression at a post-transcriptional level

By silencing transposons

By binding specific ligands with high affinity through steric interactions

By cleaving transcripts

25. Long non-coding RNAs (lncRNAs) are classified according to gene position relative to the closest protein-coding genes.

What would we call a lncRNA that is encoded from a locus between two protein-coding genes?

Intronic lncRNA

Intronic antisense lncRNA

Intergenic lncRNA

Exonic lncDNA